



Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-3630
Project Name	ComicCraft - AI Comic Story Creator
Team Lead	Sivapragasam S
Team Members	Deepak P, Lucky J, Elanthirayan S
Maximum Marks	5 Marks

Solution Architecture Diagram

Presentation / Client Layer	Application Layer	AI / Data Layer
index.html comic_preview.html export_success.html	FastAPI routes.py Input validation Workflow orchestration layout_builder.py exporters.py	Gemini Flash Gemini Pro Stable Diffusion static/panels static/exports

Architecture Flow

Browser → FastAPI routes.py → Gemini/Stable Diffusion modules → layout builder → PDF exporter → Jinja2 preview/export pages.

Core Modules

- gemini_flash.py: generate_outline()
- gemini_pro.py: generate_story()
- image_generator.py: generate_image()
- layout_builder.py: build_comic_layout()
- exporters.py: save_pdf()
- routes.py: application routes

Storage Paths

Generated panel images: static/panels. Exported PDF files: static/exports.